

State-Dependent Anonymity: Selection and State Evolution as an Attack Surface

Unified archive note v0.21 (2026-02-28; archive v145)

Abstract

Anonymity mechanisms for overlays and anonymous DHTs are increasingly *stateful*: routing and contact decisions depend on reputations, congestion signals, caches, and operator policies. State improves performance and security, but it creates a control-plane channel in which tiny per-epoch selection biases can amplify over time, and in which public state readouts (scores, load reports, cache summaries) can themselves become witnesses. We formalize two indistinguishability games—*selection privacy* and *state-evolution privacy*—and give sharp lower bounds showing how partial observation plus repetition yields near-certain identification. We then give stability/accounting upper bounds (DP-style) that translate implementable control-plane constraints into adversary advantage bounds and effective anonymity set sizes. Finally we isolate a practical mitigation pattern: *state privatization*, in which selection kernels, state releases, and monitoring are compiled into a single auditable budget. This paper is designed to be non-redundant with the prior series on stop-time padding, scheduling/compiler for metadata hiding, and spectral delegation; we cite those works when needed.

Compilation note. This paper is brief-by-design. A companion addendum collects extended calibration vignettes and worked time-to-identification conversions (State 1A).[65]

Scope. We isolate and formalize *State-Dependent Anonymity*: as a reusable contract surface in the anonymity / Anonymous-DHT series.

Units. Budgets are published in bits (use $\log_2$ and 2^ϵ); results stated in nats can be converted by dividing by $\ln 2$ (see Synthesis 8).[55]

Imports. This paper budgets *control-plane state* (scores, caches, load signals) as a witness family. We treat stop-time padding and schedule compilation as imported from the backbone wiki posts and cite them where needed. We also treat the endpoint metric interface (odds inflation and identification sizing) as imported; see Anonymity A and Synthesis 5.[58, 59] Trace-interface naming (exposure normal forms, ENF-ID) is imported from Synthesis 3, and threat/repetition windows (TW declarations, TW-ID) from Synthesis 4.[60, 61] Contract objects and publication manifests follow Synthesis 0 and the ABOM/OINL note (Anonymity C).[56, 57] For certificate/log mechanics and artifact availability, see the series artifact policy.[54] Notation conventions for keys, state, transcripts, and transcript projections follow the series notation sheet (Synthesis 8).[55]

Figure omitted in source-only archive.
 (Plot/diagram and scripts available on request.)

Figure 1: State-dependent anonymity as a control-plane feedback loop. Small per-epoch selection biases (from state) can amplify over time; public state readouts can be witnesses.

Exports. (i) Two control-plane indistinguishability games (selection privacy and state-evolution privacy) for stateful anonymous overlays and Anonymous-DHT lookups; (ii) amplification lower bounds showing how small per-epoch selection gaps become near-certain identification under repetition; and (iii) stability/accounting conditions (DP-style) that compile implementable control-plane constraints into odds-inflation / identification budgets exportable as receipts and ABOM/OINL entries.[56, 57]

Claim fields (non-negotiable). All games and bounds in this paper are parameterized by (i) a declared transcript projection (by `exposure_nf_id`, Synthesis 3) and (ii) a declared repetition/threat window (by `tw_id`, Synthesis 4).[60, 61] For stateful deployments, the public claim also carries a third claim field: a digest-bound `state_decl_id` committing to a canonical state-surface declaration (what long-lived control-plane state surfaces are in scope and what is excluded).[63, 64] This prevents “same `state_contract_id`, different state surface” drift from hiding off-surface.

Published receipts should therefore carry `exposure_nf_id`, `tw_id`, and (when applicable) `state_decl_id` as explicit line-item claim fields (Synthesis 12), and package the same ids in ABOM/OINL (Anonymity C).[62, 57] Replay-plan drift is handled analogously via `plan_spec_id`; equivalence/drift rules for both ids are centralized in Synthesis 18.[63]

Artifact policy. This archive is source-only; supporting artifacts (plots, scripts, datasets, and tooling) are available on request.[54]

1 Introduction

Stateful optimizations are now a default feature of anonymous overlays and anonymous DHT-style systems. Selection and routing decisions depend on *evolving* signals such as reputation, congestion, and caches; operators also run monitors and publish summaries. This paper treats that statefulness as a first-class anonymity channel: the *control plane* becomes a stochastic process whose visible actions and releases can leak secrets even when payloads are protected.

Control-plane loop. Figure 1 highlights the structural pattern. A secret (destination, user profile, workload class) influences hidden state, state influences selections, and selections feed back into state. An adversary need not observe payloads: partial visibility of selections, state readouts, or induced probes can suffice.

Scope and non-redundancy. We focus on *state-dependent* leakage. We treat stop-time padding and schedule equalization as imported mechanisms from the companion series [5, 1]; we use spectral delegation results when we need mixing identities [2]. For DHT-specific contact-set floors (MUCC), see AnonDHT State 2.[7] For deployment-side monitoring, alarms, and CPPC manifests, see AnonDHT State 3.[8]

Deployed control-plane configuration surfaces (e.g., delegated-routing endpoint sets, AutoConf resolution, and OHTTP/chunking feature drift) are treated separately in Synthesis 30 and should be pinned via `state_decl_id` / separation manifests rather than left implicit.[9]

Contributions.

- **Games:** we define *selection privacy* and *state-evolution privacy* for control-plane processes.
- **Lower bounds:** witness-style distinguishers and repetition/amplification theorems that quantify how tiny per-epoch gaps become near-certain identification.
- **Upper bounds:** stability/accounting conditions (DP-style) that translate implementable control-plane constraints into advantage bounds and effective anonymity set guarantees.
- **Mitigations:** a practical *state-privatization* pattern and a refresh/expiration perspective for making leakage time-limited.

2 Model and anonymity games

Parameterization by interface and window. Let ENF denote the exposure normal form that names the transcript projection revealed to the adversary, and let TW denote the repetition/threat window. We write their claim fields as `exposure_nf_id` and `tw_id` (full digests in ABOM). When we say “ n epochs” or “a horizon of n lookups”, that n is the window field of TW, and the adversary’s view is always the projection named by ENF.

Protocol as a process. Fix a secret $\sigma \in \{0, 1\}$ (or more generally $\sigma \in [K]$). A stateful protocol induces a stochastic process $(S_t, A_t, R_t)_{t \geq 1}$ where: (i) S_t is internal state (scores, queues, cache contents, committee tables), (ii) A_t is the control-plane action (selected relay/committee/contact-set, schedule choice), and (iii) R_t is any *released* state (published weights, beacons, alerts). The adversary observes $O_t = \phi(A_t, R_t)$ for some observation map ϕ capturing partial compromise, sampling, or probing.

Selection privacy. Selection privacy asks what an observer learns from the distribution of chosen control-plane actions.

Definition 1 (Selection-privacy experiment). *Let View_σ denote the distribution of the adversary transcript $O_{1:T}$ when the secret is σ . The selection advantage is*

$$\text{Adv}_T^{\text{sel}} := \text{TV}(\text{View}_0, \text{View}_1),$$

where $\text{TV}(P, Q) = \sup_E |P(E) - Q(E)|$.

This definition is agnostic to implementation: it applies whether actions are relay choices, DHT committee contacts, or schedule selections.

Remark 1 (MUCC as a marginal instantiation (DHT contact sets)). *If actions encode a (possibly multi-round) committee contact set in an anonymous DHT, then a common design goal is that the destination key does not affect which committees are contacted. Paper 2 studies a weak, marginal-only version of this objective (MUCC): each individual committee is contacted with key-independent probability [7]. MUCC does not imply selection privacy in the above sense (the joint distribution of the contacted set may still depend on the key), but it is enough to force strong availability/contact floors.*

State-evolution privacy. State-evolution privacy isolates leakage from public state readouts that drive future choices.

Definition 2 (State-evolution-privacy experiment). *Let Rel_σ be the transcript distribution when the observer sees only releases $R_{1:T}$ (and any derived metadata) under secret σ . Define*

$$\text{Adv}_T^{\text{state}} := \text{TV}(\text{Rel}_0, \text{Rel}_1).$$

Coupling view: observation as a channel. Partial observability is an additional channel applied to the underlying action/state process. This can *amplify* or *contract* distinguishability depending on what is exposed. A useful (and implementable) abstraction is to treat observation as subsampling or coarsening, which often yields privacy amplification in DP analyses. We use this later as an upper-bounding tool.

Beyond binary secrets. For $\sigma \in [K]$, the natural goal is to upper bound the maximum a-posteriori (MAP) identification probability. In Section 4, we relate stability to odds inflation and to an effective anonymity set K_{eff} .

3 Witness lower bounds and amplification

A core lesson of state-dependent anonymity is that *any* object whose selection probability differs across secrets is a witness. Repeated exposure turns tiny gaps into near-certain identification.

3.1 Single-witness distinguishers

Let W be a per-epoch event in the observer view (e.g., “committee j was contacted”, “relay r was selected”, “cache hit observed”). Let $p_\sigma := \Pr[W \mid \sigma]$.

Theorem 1 (Witness bound). *For any horizon T , the selection advantage satisfies*

$$\text{Adv}_T^{\text{sel}} \geq \text{TV}(\text{Bin}(T, p_0), \text{Bin}(T, p_1)),$$

whenever the observer can extract (possibly via a measurable function of $O_{1:T}$) an i.i.d. witness indicator sequence $(\mathbf{1}\{W_t\})_{t=1}^T$ with per-epoch probabilities p_0, p_1 .

Proof. Let g map the full transcript $O_{1:T}$ to the witness count $N := \sum_{t=1}^T \mathbf{1}\{W_t\}$. By assumption, under secret σ the induced distribution of N is $\text{Bin}(T, p_\sigma)$. Total variation distance cannot increase under post-processing (measurable maps), so

$$\text{TV}(\text{View}_0, \text{View}_1) \geq \text{TV}(g(\text{View}_0), g(\text{View}_1)) = \text{TV}(\text{Bin}(T, p_0), \text{Bin}(T, p_1)).$$

□

The i.i.d. condition holds in many “static secret” analyses (fixed destination / profile) and is a conservative approximation in slowly drifting regimes.

Figure omitted in source-only archive.
 (Plot/diagram and scripts available on request.)

Figure 2: Illustrative sample complexity scaling: tiny per-epoch witness gaps require long horizons, but eventually yield near-certain identification.

3.2 Repetition and sample complexity

Write $\Delta := |p_1 - p_0|$. Testing two Bernoulli means with a sequential probability ratio test (SPRT) yields sharp amplification rates [25, 26].

Theorem 2 (Repeated-witness amplification). *Assume witness indicators $X_1, \dots, X_T$ are i.i.d. Bernoulli(p_σ) under secret $\sigma \in \{0, 1\}$. There exists a test with total error at most η whenever*

$$T \gtrsim \frac{\log_2(1/\eta)}{\Delta^2}.$$

Conversely, for Δ small and p_σ bounded away from 0, 1, any test needs $T = \Omega(\log_2(1/\eta)/\Delta^2)$ samples.

Figure 2 gives a “numbers-first” view of this scaling.

Calibration snapshot (informal). Table 1 gives a compact numbers-first view of the $1/\Delta^2$ amplification scaling in Theorem 2. The table is illustrative (independence/nonstationarity ignored) and exists to make the quadratic dependence operationally visible.

Per-epoch gap Δ	Target total error η	Rule-of-thumb T	
0.01	0.05	$\approx 6 \times 10^4$	“tiny bias” needs $\sim$ weeks at minute cadence (or day
0.02	0.05	$\approx 1.5 \times 10^4$	$\sim$ 10 days at minute cadence (or $\sim$ 2 months
0.05	0.05	$\approx 2.4 \times 10^3$	$\sim$ 40 minutes at 1s cadence (or $\sim$ 7 hour
0.10	0.05	$\approx 6 \times 10^2$	$\sim$ 10 minutes at 1s cadence (or $\sim$ 1.2 hour

Table 1: Back-of-the-envelope calibration for Theorem 2 using $T \approx 2\log_2(1/\eta)/\Delta^2$. These are illustrative and ignore dependence and nonstationarity; they serve only to show the strong $1/\Delta^2$ scaling.

3.3 Adaptive witnesses and many streams

Real deployments generate many potential witnesses (keys, committees, relays, cache lines, alarms). Even when any *single* stream is weak, scanning or adaptively choosing witnesses can drive identification.

Corollary 1 (Searching N witness streams adds a $\log_2 N$ factor). *Suppose the observer can compute N candidate witness streams $\{X_t^{(i)}\}_{t \geq 1}$ for $i \in [N]$, where each $X_t^{(i)} \in \{0, 1\}$ and (under a fixed secret $\sigma \in \{0, 1\}$) each stream is i.i.d. Bernoulli($p_{\sigma,i}$). Let $\Delta_\star := \max_i |p_{1,i} - p_{0,i}|$. Then there exists an anytime-valid sequential test that (i) adaptively chooses a stopping time τ , (ii) has total error at most η , and (iii) achieves the scaling*

$$\tau = O\left(\frac{\log_2(N/\eta)}{\Delta_\star^2}\right)$$

up to constant factors (in regimes where $p_{\sigma,i}$ are bounded away from 0, 1).

Proof sketch. Run a likelihood-ratio e-process (SPRT-style) on each candidate stream in parallel and stop when any crosses a calibrated threshold. Ville’s inequality gives time-uniform control under optional stopping, and the extra $\log_2 N$ factor comes from the multiple-testing calibration over N streams; see the SAVI/e-value framework [27, 39]. $\square$

We avoid duplicating the full many-testing development in the companion series paper [6]. Instead, Paper 3 in this archive uses e-values/anytime-valid monitoring as an *accounting interface* for many public streams (alerts, beacons, summaries).

4 Stability/accounting upper bounds

Lower bounds say what can go wrong. Upper bounds should be implementable: they must correspond to constraints engineers can enforce on selection kernels, releases, and monitors. A natural language is *stability*: small changes in the secret should not strongly tilt control-plane probabilities.

4.1 DP-style stability for control planes

Definition 3 (Per-epoch max-divergence stability). *A per-epoch observation distribution Obs_σ satisfies (ε, δ) stability if for all measurable E ,*

$$\Pr[\text{Obs}_0 \in E] \leq 2^\varepsilon \Pr[\text{Obs}_1 \in E] + \delta \quad \text{and} \quad \Pr[\text{Obs}_1 \in E] \leq 2^\varepsilon \Pr[\text{Obs}_0 \in E] + \delta.$$

This is the standard (approximate) differential privacy inequality [15] applied to control-plane views. It immediately yields a TV bound and composes over time.

Proposition 1 (Stability implies TV and odds bounds). *If Obs_σ satisfy (ε, δ) stability, then*

$$\text{TV}(\text{Obs}_0, \text{Obs}_1) \leq \min\{1, 2^\varepsilon - 1 + \delta\}.$$

If $\delta = 0$ and the distributions admit densities with respect to a common measure, then for any realized observation o ,

$$2^{-\varepsilon} \leq \frac{d\text{Obs}_0}{d\text{Obs}_1}(o) \leq 2^\varepsilon,$$

so posterior odds between $\sigma = 0$ and $\sigma = 1$ inflate by at most 2^ε in either direction. Under basic composition over epochs with per-epoch $(\varepsilon_t, \delta_t)$, the transcript satisfies $(\sum_t \varepsilon_t, \sum_t \delta_t)$, and the same bounds apply with $\varepsilon = \sum_t \varepsilon_t$ and $\delta = \sum_t \delta_t$.

Proof sketch. Let E attain the TV supremum so that $\text{TV}(P, Q) = P(E) - Q(E)$ for $P = \text{Obs}_0, Q = \text{Obs}_1$. Apply the stability inequality to E and to its complement and rearrange. The odds statement for $\delta = 0$ is the pointwise likelihood-ratio form of max-divergence stability.

4.2 Stopping-safe drift audits (optional stopping)

Control planes are monitored in real time: operators run drift detectors on selections, scores, and release channels, and trigger mitigation when the detector fires. But the trigger time is *adaptive* (a stopping time), and naive “check every epoch” thresholds can severely understate false-alarm rates. A clean and receipt-friendly fix is to publish drift evidence as a *time-uniform* bound derived from a nonnegative supermartingale (an *e-value*).[34, 35]

Lemma 1 (Ville inequality / time-uniform validity). *Let $(E_t)_{t \geq 0}$ be a nonnegative supermartingale with $E_0 = 1$ under a null model (e.g., “no drift”), and let τ be any (possibly data-dependent) stopping time. Then for any $\alpha \in (0, 1)$,*

$$\Pr \left[\sup_{t \geq 0} E_t \geq 1/\alpha \right] \leq \alpha, \quad \text{and hence} \quad \Pr [E_\tau \geq 1/\alpha] \leq \alpha.$$

Remark 2 (Receipt hook). *A drift monitor can therefore publish: (i) the detector family (as part of a state declaration), (ii) the chosen α , and (iii) a digest-bound evidence object containing the realized E_t trace up to τ . Because the guarantee is time-uniform, auditors do not need to know how often the monitor checked or how the trigger time was chosen. This is orthogonal to the privacy bound itself: if a triggered mitigation changes the control-plane kernel, then the updated receipt should roll the relevant envelope ids (new `state_decl_id` and/or new knobs), and any adaptive accounting should be expressed via conditional certificates (Synthesis 21).[66, 63]*

□

4.3 Concrete kernel: softmax stability from bounded score sensitivity

Many control planes implement selection via a scored softmax (or, equivalently, the exponential mechanism [16]). Let scores $x_i(\sigma) \in \mathbb{R}$ be computed for each option $i \in [M]$, and select $I \in [M]$ with

$$\Pr[I = i \mid \sigma] = \frac{\exp(x_i(\sigma)/\tau)}{\sum_{j=1}^M \exp(x_j(\sigma)/\tau)},$$

where $\tau > 0$ is a temperature.

Lemma 2 (Softmax max-divergence bound). *Assume $\|x(\sigma=0) - x(\sigma=1)\|_\infty \leq \Delta$. Then the selection distribution satisfies $(\varepsilon, 0)$ stability with*

$$\varepsilon = \frac{2\Delta}{\tau}.$$

Proof. For each i , the numerator ratio is bounded by $\exp(\Delta/\tau)$, while the normalizer ratio is bounded by $\exp(\Delta/\tau)$ in the opposite direction:

$$\frac{\sum_j \exp(x_j(1)/\tau)}{\sum_j \exp(x_j(0)/\tau)} \leq \exp(\Delta/\tau).$$

Thus $\Pr[I = i \mid 0] / \Pr[I = i \mid 1] \leq \exp(2\Delta/\tau)$ for every i , yielding the max-divergence form of $(\varepsilon, 0)$ stability. □

Lemma 2 turns an engineering control (score sensitivity Δ and temperature τ) into an explicit per-epoch stability budget. In practice, Δ can be enforced by clipping or smoothing state-dependent scores before selection, and by limiting how much any single secret-dependent update can move a score.

Partial observation as privacy amplification. When the observer sees only a subsample or coarsening of actions, stability can improve. We treat this using privacy amplification by subsampling [46] as a conservative upper bound.

Figure omitted in source-only archive.
 (Plot/diagram and scripts available on request.)

Figure 3: Example calibration between ε -style stability and effective anonymity size K_{eff} .

4.4 Composition with adaptive parameters

In stateful systems, selection parameters and monitoring intensity may be chosen adaptively. Stability budgets must therefore be tracked with an *online accountant*. Privacy odometers/filters provide a canonical interface for this pay-as-you-go regime [19, 41]. When multiple mechanisms interleave (e.g., routing + monitoring + retries), interactive concurrent composition is relevant [44, 42]. We defer the deployment-side discussion (budget beacons, alarm release, and practical filter choices) to Paper 3 [8].

4.5 Effective anonymity sets from odds inflation

For K profiles, a compact operational summary is an *effective anonymity set*. A simple and widely useful interpretation is via worst-case odds inflation. If stability guarantees that the likelihood ratio between any two profiles is at most ρ (over the observed horizon), then MAP identification is at most ρ/K under a uniform prior, suggesting $K_{\text{eff}} \approx K/\rho$. Figure 3 shows a typical calibration between an ε -style budget and K_{eff} .

4.6 Expiration and refresh

If a system mixes and periodically refreshes state (or applies gradual privacy expiration), then leakage can become time-limited. We treat this as “expiration-aware” accounting and cite discounted DP and gradual expiration mechanisms [29, 30].

4.7 Contractive state updates justify expiration-style models

Many practical control-plane states are implemented with exponential forgetting (EWMA reputations, cache decay, moving averages). This yields a clean contraction bound that turns an “infinite horizon” into an *effective* finite one.

Lemma 3 (EWMA contraction). *Let a scalar state $s_t \in \mathbb{R}$ evolve as*

$$s_{t+1} = (1 - \lambda)s_t + \lambda u_t, \quad \lambda \in (0, 1],$$

where u_t is an arbitrary input sequence. For any two executions with the same inputs but different initial states s_0, s'_0 , the difference contracts geometrically:

$$|s_t - s'_t| \leq (1 - \lambda)^t |s_0 - s'_0|.$$

More generally, if the inputs differ by at most $|u_t - u'_t| \leq \Delta_u$ for all t , then

$$|s_t - s'_t| \leq (1 - \lambda)^t |s_0 - s'_0| + \Delta_u \sum_{i=0}^{t-1} (1 - \lambda)^i \leq (1 - \lambda)^t |s_0 - s'_0| + \Delta_u / \lambda.$$

Proof. Unroll the recursion and sum the resulting geometric series. □

Figure omitted in source-only archive.
(Plot/diagram and scripts available on request.)

Figure 4: Illustrative decay/expiration curve for stateful leakage.

Figure omitted in source-only archive.
(Plot/diagram and scripts available on request.)

Figure 5: Example of inhomogeneous spectral leakage.

Remark 3 (Vector-state variant). *If the state is vector-valued $s_t \in \mathbb{R}^p$ and evolves via the same EWMA recursion $s_{t+1} = (1 - \lambda)s_t + \lambda u_t$, then the same geometric contraction holds in any norm: $\|s_t - s'_t\| \leq (1 - \lambda)^t \|s_0 - s'_0\|$ (plus the corresponding input-difference term). If a selection kernel is Lipschitz in state in the sense that $\text{TV}(K(s), K(s')) \leq L\|s - s'\|$ (or via an SDPI/Dobrushin coefficient for the induced channel), then contraction yields an explicit “effective memory” bound that is directly usable in expiration-aware accounting.*

If a selection kernel $K(\cdot)$ is Lipschitz in state (e.g., $\text{TV}(K(s), K(s')) \leq L\|s - s'\|$ or a comparable max-divergence notion), then Lemma 3 implies that state differences from ancient epochs are exponentially downweighted. Operationally, $1/\lambda$ acts as an “effective memory” in epochs, aligning with discounted/gradual-expiration accounting.

5 Mitigation pattern: state privatization

A practical lesson is that control-plane choices, public releases, and monitoring must be designed *together*. Treating them independently tends to create accidental witnesses.

State privatization compiler. We advocate a compiler viewpoint: (i) define a target selection kernel family, (ii) define which state is released and at what cadence, (iii) define monitoring/alerts, and (iv) attach a single budget/accountant to the whole control plane. Paper 3 in this archive develops a deployment pattern for this idea (measurement firewall + CPPC manifests).

Refresh to make leakage expire. If the state process mixes and occasionally refreshes, then older secrets have diminishing influence on future actions. Figure 4 illustrates an “identification half-life” viewpoint.

Caution: inhomogeneous mixing can leak. Not all mixing is benign: non-homogeneous dynamics can create spectral leakage modes. We rely on the series’ spectral delegation paper for detailed identities [2]; Figure 5 is a warning example.

Checklist. In practice, the following questions catch most state-dependent failures: (i) what state influences selection, and who can observe it; (ii) which public releases can become probes/oracles; (iii) how are retries/abort patterns equalized (see stop-time series work [5]); (iv) which monitors/alerts exist, and are they closed-view; (v) what refresh mechanisms bound the influence horizon.

6 Related work

Statefulness pervades modern anonymity. Tor’s design emphasizes practical deployability [12] and has a long line of work on congestion/selection. Recent bandwidth-measurement manipulation attacks (e.g., TorMult and MirageFlow) illustrate that published selection weights can be steered, turning the control plane into an oracle for downstream traffic analysis [49, 48]. DPSelect is an early proposal to apply max-divergence accounting directly to Tor guard selection [50].

In incentivized mixnets, routing and rewards are coupled to stake/performance; reputation becomes a direct attack surface. Recent work analyzes and attacks Nym’s reputation system (Cao and Green) [10]. The PETS 2026 paper on website fingerprinting on Nym released artifacts/datasets on Zenodo [52]. Flow-correlation and attribution attacks continue to improve (including deep learning) [32, 33, 40]. For mixnets specifically, MixMatch demonstrates flow matching at the mixnet level [14]. For Tor onion services, sliding subset-sum correlation is a recent example of long-horizon correlation under partial observability [36].

Our upper bounds borrow heavily from differential privacy as a stability language [15]. We rely on odometers/filters for adaptive parameter regimes [19] and on interactive/concurrent composition results when multiple deployed mechanisms interleave [44, 42]. We use privacy amplification by subsampling as a conservative model of partial observation [46]. Repeated aggregate releases over time are treated in the continual-observation DP literature [17, 18]. Always-valid monitoring and e-values provide a clean technical interface between detection and accounting [26, 27]. We develop the deployment-side implications (including multiple-testing and public-alert leakage) in Paper 3.

For explicitly stateful secrets and correlated background knowledge, Pufferfish provides a general Bayesian framework that treats (i) the protected secrets, (ii) discriminative secret pairs, and (iii) data evolution scenarios as first-class inputs [47]. While we keep the present paper in a lightweight max-divergence/DP style, Pufferfish-style instantiations are a natural way to formalize state-evolution privacy when the operator and observer share an explicit model of state dynamics.

Caching as a privacy side channel has a long history (timing/oracle effects) and appears in anonymity, content routing, and DHTs. For anonymous DHTs, the control-plane state includes routing tables, bucket replacement/refresh, and provider-record lifetimes. Paper 2 of this split archive treats DHT-specific anonymous lookup mechanisms, contact-set witness floors (MUCC), and deployed-scale context (including recent IPFS evidence).

7 Takeaways

- **State is an anonymity surface:** any key-dependent influence on selection weights, cache contents, or public scores is a witness stream and must be budgeted.
- **Small gaps amplify fast:** repetition turns per-epoch gaps Δ into near-certain identification on the $1/\Delta^2$ scale; refresh horizons and contractive updates are operationally decisive.
- **Stability is the right upper-bound language:** DP-style max-divergence bounds on the control-plane kernel export directly to odds-inflation and effective anonymity-set statements used elsewhere in this series.
- **Monitoring must be closed-view:** publishable alarms and dashboards are themselves releases; prefer closed-view auditing (AnonDHT State 3) and commit to what is released via receipts (Release A).

References

- [1] Anonymous. *Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups*. Series paper (published), 2026. Wiki: [[2026.01.22 - Mathematics: Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups]].
- [2] Anonymous. *Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays*. Series paper (published), 2026. Wiki: [[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]].
- [3] Anonymous. *Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding*. Series paper (published), 2026. Wiki: [[2026.01.25 - Mathematics: Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding]].
- [4] Anonymous. *Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy*. Series paper (published), 2026. Wiki: [[2026.01.29 - Mathematics: Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy]].
- [5] Anonymous. *Stop-Time Padding Addendum (unique-only): Max-Mixture Separations and Tail-Sign Deadline Normal Forms*. Series note (published), 2026. Wiki: [[2026.01.26 - Mathematics: Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms]].
- [6] Anonymous. *Many-Testing Lower Bounds for Low-Latency Privacy on Hierarchical Overlays*. Series paper (published), 2026.
- [7] Anonymous. Committee Contact-Set Privacy in Anonymous DHT Lookups: MUCC, Availability Floors, and Abort-Safe Equalization. Companion preprint in this archive (AnonDHT State 2).
- [8] Anonymous. Closed-View Auditing for Stateful Anonymity: Odometers, Alarms, and Machine-Checkable CPPCs. Companion preprint in this archive (AnonDHT State 3).
- [9] Anonymous. Deployed Observation Surfaces for Anonymous DHTs: Control-plane knobs in IPFS/libp2p delegated routing and OHTTP evolution. Series synthesis note (Synthesis 30), 2026. (This archive.)
- [10] Xinmu Alexis Cao and Matthew Green. Analysis and Attacks on the Reputation System of Nym. Proceedings on Privacy Enhancing Technologies (PoPETs), 2026 (Issue 2). Cryptology ePrint Archive, Report 2026/101. <https://eprint.iacr.org/2026/101> Author page (links to artifact): <https://alexiscaogithub.io/>
- [11] David Chaum. Untraceable Electronic Mail, Return Addresses, and Digital Pseudonyms. Communications of the ACM, 24(2), 84–90, 1981.
- [12] Roger Dingledine and Nick Mathewson and Paul Syverson. Tor: The Second-Generation Onion Router. Proceedings of the 13th USENIX Security Symposium, 2004.
- [13] Ania M. Piotrowska and Jamie Hayes and Tariq Elahi and Sebastian Meiser and George Danezis. The Loopix Anonymity System. USENIX Security Symposium, 2017.
- [14] Lennart Oldenburg and Marc Juarez and Enrique Argones Rúa and Claudia Diaz. MixMatch: Flow Matching for Mixnet Traffic. Proceedings on Privacy Enhancing Technologies, 2024(2), 276–294, 2024. DOI: 10.56553/popets-2024-0050. <https://petsymposium.org/popets/2024/popets-2024-0050.pdf>
- [15] Cynthia Dwork and Aaron Roth. The Algorithmic Foundations of Differential Privacy. Foundations and Trends in Theoretical Computer Science, 9(3–4), 211–407, 2014.

- [16] Frank McSherry and Kunal Talwar. Mechanism Design via Differential Privacy. 48th Annual IEEE Symposium on Foundations of Computer Science (FOCS), 94–103, 2007. DOI: 10.1109/FOCS.2007.41. Introduces the exponential mechanism viewpoint used widely for DP-style selection. <https://kunalatalwar.org/papers/expmech.pdf>
- [17] Cynthia Dwork and Moni Naor and Toniann Pitassi and Guy N. Rothblum. Differential Privacy Under Continual Observation. Proceedings of the 42nd ACM Symposium on Theory of Computing (STOC), 715–724, 2010. DOI: 10.1145/1806689.1806787. <https://dl.acm.org/doi/10.1145/1806689.1806787>
- [18] T.-H. Hubert Chan and Elaine Shi and Dawn Song. Private and Continual Release of Statistics. International Colloquium on Automata, Languages, and Programming (ICALP), Part II, 6199, 405–417, 2010. Also available as IACR ePrint 2010/076. <https://people.eecs.berkeley.edu/~dawnsong/papers/2010%20private%20continual.pdf>
- [19] Ryan Rogers and Aaron Roth and Jonathan Ullman and Salil Vadhan. Privacy Odometers and Filters: Pay-as-you-Go Composition. Advances in Neural Information Processing Systems (NeurIPS), 2016.
- [20] Mathias Lécuyer. Practical Privacy Filters and Odometers with Rényi Differential Privacy and Applications to Differentially Private Deep Learning. 2021. arXiv:2103.01379
- [21] Ilya Mironov. Rényi Differential Privacy. 2017 IEEE 30th Computer Security Foundations Symposium (CSF), 2017. Also available as arXiv:1702.07476
- [22] Borja Balle and Gilles Barthe and Marco Gaboardi and Justin Hsu and Tetsuya Sato. Hypothesis Testing Interpretations and Rényi Differential Privacy. International Conference on Artificial Intelligence and Statistics (AISTATS), 2020.
- [23] Peter Kairouz and Sewoong Oh and Pramod Viswanath. The Composition Theorem for Differential Privacy. International Conference on Machine Learning (ICML), 2015. Also available as arXiv:1311.0776
- [24] Tianhao Chan and Elaine Shi and Dawn Song. Private and Continual Release of Statistics. ACM Transactions on Information and System Security, 2011. Early conference version in 2010.
- [25] Abraham Wald. Sequential Tests of Statistical Hypotheses. Annals of Mathematical Statistics, 16(2), 117–186, 1945.
- [26] Steven R. Howard and Aaditya Ramdas and Jon McAuliffe and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. Annals of Statistics, 49(2), 1055–1080, 2021. DOI: 10.1214/20-AOS1991. Also available as arXiv:1810.08240. <https://projecteuclid.org/journals/annals-of-statistics/volume-49/issue-2/Time-uniform-nonparametric-nonasymptotic-confidence-sequences/10.1214/20-AOS1991.pdf>
- [27] Aaditya Ramdas and Ruodu Wang. Hypothesis Testing with E-values. Foundations and Trends® in Statistics, 1(1–2), 1–390, 2025. DOI: 10.1561/36000000002. Also available as arXiv:2410.23614.
- [28] Aaditya Ramdas and Peter Grünwald and Vladimir Vovk and Glenn Shafer. Game-Theoretic Statistics and Safe Anytime-Valid Inference. 2023. arXiv:2210.01948
- [29] Farhad Farokhi. Temporally Discounted Differential Privacy for Evolving Datasets on an Infinite Horizon. Proceedings of the 2020 ACM/IEEE 11th International Conference on Cyber-Physical Systems (ICCPs), 2020. DOI: 10.1109/ICCPs48487.2020.00008. Conference version with DOI; open preprint on arXiv. <https://arxiv.org/abs/1908.03995>
- [30] Joel Daniel Andersson and Monika Henzinger and Rasmus Pagh and Teresa Anna Steiner and Jalaj Upadhyay. Continual Counting with Gradual Privacy Expiration. Advances in Neural Information Processing Systems (NeurIPS), 2024. DOI: 10.48550/arXiv.2406.03802. NeurIPS 2024; see also NeurIPS proceedings entry (ACM DL: 10.5555/3737916.3738215). <https://arxiv.org/abs/2406.03802>

- [31] Ziyu Xu and Aaditya Ramdas. Online multiple testing with e-values. Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS), 238, 2024. PMLR 238.
- [32] Milad Nasr and Alireza Bahramali and Amir Houmansadr. DeepCorr: Strong Flow Correlation Attacks on Tor Using Deep Learning. Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security (CCS), 2018. DOI: 10.1145/3243734.3243824. Also available as arXiv:1808.07285.
- [33] Se Eun Oh and Taiji Yang and Nate Mathews and James K. Holland and Mohammad Saidur Rahman and Nicholas Hopper and Matthew Wright. DeepCoFFEA: Improved Flow Correlation Attacks on Tor via Metric Learning and Amplification. Proceedings of the IEEE Symposium on Security and Privacy (SP), 1915–1932, 2022. DOI: 10.1109/SP46214.2022.9833801.
- [34] Howard, Steven R. and Ramdas, Aaditya and McAuliffe, Jon and Sekhon, Jasjeet. Time-uniform Chernoff bounds via nonnegative supermartingales. 2018. arXiv:1808.03204
- [35] Wang, Ruodu and Ramdas, Aaditya. False discovery rate control with e-values. 2020. arXiv:2009.02824 (published in JRSSB 2022)
- [36] Daniela Lopes and Jin-Dong Dong and Pedro Medeiros and Daniel Castro and Diogo Barradas and Bernardo Portela and João Vinagre and Bernardo Ferreira and Nicolas Christin and Nuno Santos. Flow Correlation Attacks on Tor Onion Service Sessions with Sliding Subset Sum. Network and Distributed System Security Symposium (NDSS), 2024. DOI: 10.14722/ndss.2024.24337. <https://www.ndss-symposium.org/ndss-paper/flow-correlation-attacks-on-tor-onion-service-sessions-with-sliding-subset-sum/>
- [37] Ruodu Wang and Aaditya Ramdas. False discovery rate control with e-values. Journal of the Royal Statistical Society: Series B, 84(3), 2022. doi:10.1111/rssb.12489 (preprint arXiv:2009.02824)
- [38] Lasse Fischer and Ziyu Xu and Aaditya Ramdas. An online generalization of the (e-)Benjamini–Hochberg procedure. 2024. arXiv:2407.20683
- [39] Wang, Hongjian and Dandapanthula, Sanjit and Ramdas, Aaditya. Anytime-valid FDR control with the stopped e-BH procedure. Statistics & Probability Letters, 226, 110512, 2025. DOI: 10.1016/j.spl.2025.110512. Preprint: arXiv:2502.08539.
- [40] Gurjot Singh and Alim Dhanani and Diogo Barradas. RevealNet: Distributed Traffic Correlation for Attack Attribution on Programmable Networks. 2025. arXiv:2505.00618 (accepted to IEEE NCA 2025)
- [41] Justin Whitehouse and Aaditya Ramdas and Ryan Rogers and Steven Wu. Fully-Adaptive Composition in Differential Privacy. Proceedings of the 40th International Conference on Machine Learning (ICML), 202, 36990–37007, 2023. ICML 2023 / PMLR; an open PDF is also mirrored by the authors. <https://proceedings.mlr.press/v202/whitehouse23a.html>
- [42] Haney, Samuel and Shoemate, Michael and Tian, Grace and Vadhan, Salil and Vyrros, Andrew and Xu, Vicki and Zhang, Wanrong. Concurrent Composition for Interactive Differential Privacy with Adaptive Privacy-Loss Parameters. Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security (CCS '23), 2023. DOI: 10.1145/3576915.3623128. Also available as arXiv:2309.05901.
- [43] Christopher Blier-Wong and Ruodu Wang. Improved thresholds for e-values. 2026. DOI: 10.48550/arXiv.2408.11307. Updated preprint (Feb 7, 2026); see also earlier arXiv versions. <https://arxiv.org/abs/2408.11307>
- [44] Salil Vadhan and Tianhao Wang. Concurrent Composition of Differential Privacy. 2021. arXiv:2105.14427
- [45] Xuechen Lyu and Zhanqiu Zhang and Ziyu Zhang and Yu-Xiang Wang. Composition Theorems for Interactive Differential Privacy. Advances in Neural Information Processing Systems (NeurIPS), 2022. neurips.cc; 2022

- [46] Borja Balle and Gilles Barthe and Marco Gaboardi. Privacy Amplification by Subsampling: Tight Analyses via Couplings and Divergences. *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. NeurIPS 2018; also available as arXiv:1807.01647
- [47] Daniel Kifer and Ashwin Machanavajjhala. Pufferfish: A Framework for Mathematical Privacy Definitions. *ACM Transactions on Database Systems*, 39(1), 1–36, 2014. DOI: 10.1145/2514689. TODS 39(1), Article 3; open author PDF. https://users.cs.duke.edu/~ashwin/pubs/pufferfish_TODS.pdf
- [48] Christoph Sendner and Jasper Stang and Alexandra Dmitrienko and Raveen Wijewickrama and Murtuza Jadliwala. MirageFlow: A New Bandwidth Inflation Attack on Tor. *Network and Distributed System Security Symposium (NDSS)*, 2024. NDSS 2024; PDF also mirrored by anonbib/freehaven. <https://www.ndss-symposium.org/ndss-paper/mirageflow-a-new-bandwidth-inflation-attack-on-tor/>
- [49] Christoph Sendner and Jasper Stang and Alexandra Dmitrienko and Raveen Wijewickrama and Murtuza Jadliwala. TorMult: Introducing a Novel Tor Bandwidth Inflation Attack. 2023. DOI: 10.48550/arXiv.2307.08550. arXiv:2307.08550 <https://arxiv.org/abs/2307.08550>
- [50] Hans Hanley and Yixin Sun and Sameer Wagh and Prateek Mittal. DPSelect: A Differential Privacy Based Guard Relay Selection Algorithm for Tor. *Proceedings on Privacy Enhancing Technologies*, 2019(2), 166–186, 2019. DOI: 10.2478/popets-2019-0025. PoPETs 2019 (open access via PETS). <https://petsymposium.org/popets/2019/popets-2019-0025.php>
- [51] Privacy Enhancing Technologies Symposium (PETS). PETS 2026 Accepted Papers List. 2026. Accepted paper list; includes “Analysis and Attacks on the Reputation System of Nym” by Xinmu Alexis Cao and Matthew Green. <https://petsymposium.org/2026/paperlist.php>
- [52] Eric Jollès. Website Fingerprinting on Nym: Attacks and Defenses, Artifact Test Data. Zenodo dataset, 2025. DOI: 10.5281/zenodo.17867461. Published 2025-12-09; artifact dataset for the PETS 2026 paper “Website Fingerprinting on Nym: Attacks and Defenses”. <https://zenodo.org/records/17867461>
- [53] Diaz, Claudia and Halpin, Harry and Kiayias, Aggelos. Reward Sharing for Mixnets. 2022. Nym cryptoeconomic paper; notes “720 hourly epochs per monthly interval” while cautioning epoch/interval lengths may change. <https://nym.com/nym-cryptoecon-paper.pdf>
- [54] Anonymous. Artifact and Disclosure Policy for the One-True-Archive (Source-Only Public Release). Synthesis 6, The-One-True-Archive (2026).
- [55] Anonymous. Notation and Minimal Model for the One-True-Archive. Synthesis 8, The-One-True-Archive (2026).
- [56] Anonymous. Contract Objects for Anonymous-DHT Privacy Budgets and Claims. Synthesis 0, The-One-True-Archive (2026).
- [57] Anonymous. ABOM and OINL for Anonymity Claims. Anonymity series Paper C, The-One-True-Archive (2026).
- [58] Anonymous. Odds Inflation, Privacy Loss, and Endpoint Anonymity Bounds. Anonymity series Paper A, The-One-True-Archive (2026).
- [59] Anonymous. Endpoint Metrics Bridge: Odds Inflation, PML/MaxL, and Identification Sizing. Synthesis 5, The-One-True-Archive (2026).
- [60] Anonymous. Trace Interfaces for Anonymous DHTs. Synthesis 3, The-One-True-Archive (2026).
- [61] Anonymous. Threat Models and Repetition Windows for Anonymous DHTs. Synthesis 4, The-One-True-Archive (2026).
- [62] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Synthesis 12, The-One-True-Archive (2026).

- [63] Anonymous. Digest-Bound Replay Plans and State Declarations: Equivalence, Minimality, and Change Control. Synthesis 18, The-One-True-Archive (2026).
- [64] Anonymous. Worked Example: Instantiating a Tiered Reader-Privacy Receipt. Synthesis 17, The-One-True-Archive (2026).
- [65] Anonymous. State-Dependent Anonymity (Addendum): Calibration Vignettes and Time-to-Identification Conversions. AnonDHT State series Paper 1A, The-One-True-Archive (2026).
- [66] Anonymous. Conditional Per-Step Budget Certificates. Series synthesis note (Synthesis 21), 2026. (This archive.)